

Garv Aggarwal

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Backend Engineer

Backend engineer with **end-to-end ownership** across distributed Java services — designing cross-service contracts, driving architecture decisions on messaging and configuration, and maintaining production reliability through on-call.

EDUCATION

Maharaja Agrasen Institute of Technology, New Delhi
B.Tech — Computer Science & Engineering (Minor: AI/ML)

CGPA: 9.0
2021 – 2025

EXPERIENCE

- Octro Inc — Member of Technical Staff (Backend Engineer)** *(promoted from intern, Mar-Jul 2025) | Aug 2025 – Present*
- Designed and operated **Java Spring Boot** microservices for a high-traffic consumer platform — shipping **7 features end-to-end** across WebSockets, RabbitMQ, Redis, MongoDB, and ZooKeeper, with full cross-service ownership and on-call responsibility.
 - Leveraged a **ZooKeeper-backed runtime config and feature-toggle system** to push settings to all **54 service instances without redeployment**, enabling zero-downtime rollouts and safer staged releases.
 - Re-architected a **blocking notification call on the auth path** into a fully async **Amazon SQS pipeline** — owned the delivery design: per-platform routing, configurable ZooKeeper thresholds, dead-letter queue, CloudWatch alerting.
 - Designed a **reward lifecycle management system across 3 microservices** — owned the full contract: grant/expiry semantics, scheduled delivery integration, and balance integrity on account merge and deletion.
 - Designed a **multi-provider OAuth account linking system** spanning 3 microservices — time-bound tokens, email verification, atomic merge with idempotency, same-user, and partial-failure guards across service boundaries.
 - Investigated and resolved **6 production incidents on-call: a distributed-lock thread-affinity violation** (`IllegalMonitorStateException`) cascading 5xx across 3 services; a WebSocket race causing duplicate session authorization; and a `NullPointerException` blocking reward delivery for active users.
 - Eliminated N+1 patterns:** batched Redis reads on a high-frequency polling service (every 10s per user) and MongoDB fan-out on the login path. Fixed Maven dependency scoping, **halving the deployed artifact size**.

StepOut — Software Developer Intern | Nov 2024 – Feb 2025

- Built a **Flask video-processing pipeline** — ingested match footage, ran per-frame **YOLO object detection** to identify and annotate players by jersey number, and re-encoded processed frames into a final output video.
- Implemented **asynchronous background upload/download** for large media files for uninterrupted transfers, and reduced redundant API calls on data-heavy screens by improving client state synchronization.

PROJECTS

Code Execution Engine | Go, Docker API, Goroutines, SSE

- Built a LeetCode-style execution engine in Go — **worker-pool architecture** (goroutines + buffered channels) supporting **500+ concurrent submissions**; each runs in an isolated Docker container with resource limits and context-based timeouts, cutting cold-start from **~3.2s to ~450ms**.
- Streamed real-time build output over **Server-Sent Events**, removing polling and cutting client-server bandwidth by **~70%**.

HTTP Load Balancer | Go, sync.RWMutex, fsnotify, YAML

- Built a **production-style HTTP load balancer** in Go — multiple routing strategies, concurrency-safe request path, **zero-downtime config reloads** via fsnotify, and active health checks with failure isolation.

TECHNICAL SKILLS

System Design: Distributed Systems, Cross-service Contracts, Event-Driven Architecture, Zero-Downtime Releases

Languages: Java, Go, JavaScript, TypeScript

Backend: Spring Boot, Microservices, REST, WebSockets

Distributed Systems & Data: Redis, ZooKeeper, RabbitMQ, Amazon SQS, MongoDB, Kafka

Cloud & DevOps: AWS (Lambda, EC2, S3, CloudWatch), Docker, Kubernetes, CI/CD, Linux